



## Perimeter of composite shapes

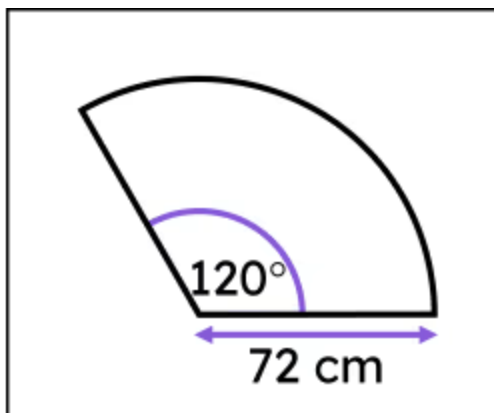
1 A quarter-circular sector is cut out of a circle. The original circle had a perimeter of  $60\pi$  cm. What is the arc length of this sector? Tick **1** correct answer

- ☒  $15\pi$  cm
- ☐  $15\pi + 30$  cm
- ☐  $15\pi + 60$  cm

2 A semicircular sector is cut out of a circle. The radius of the original circle is 31 cm, and its circumference is 194.8 cm (rounded to 1 decimal place). What is the perimeter of the sector? Tick **2** correct answers

- ☐  $31\pi$
- ☐ 97.4 cm
- ☐  $97.4 + 31$  cm
- ☒  $97.4 + 62$  cm
- ☒ 159.4 cm

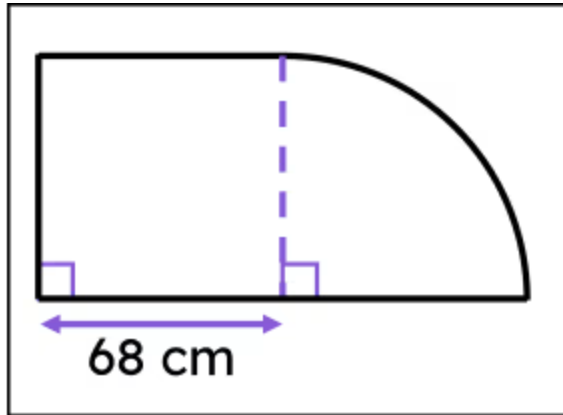
3 Calculate the perimeter of this sector, giving your answer in cm and rounding your answer to 1 decimal place. Fill in the blank



294.8



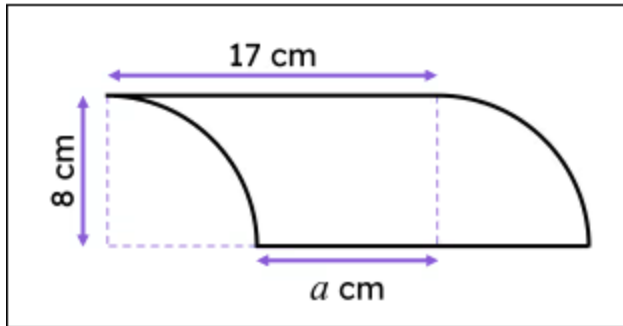
- 4 This composite shape is composed of a circular sector and a square. Which of these statements are true about the perimeter of this composite shape that has been broken into component lengths. Tick **2** correct answers



- ☐ There are three line segments of 68 cm.
- ☒ There are four line segments of 68 cm.
- ☐ There are five line segments of 68 cm.
- ☐ The length of the arc is  $68\pi$  cm.
- ☒ The length of the arc is  $34\pi$  cm.



- 5 This composite shape is composed of two quarter-circles and a rectangle. Match the statement about the component parts of the composite shape's perimeter to the correct value. Write the correct letter in each box

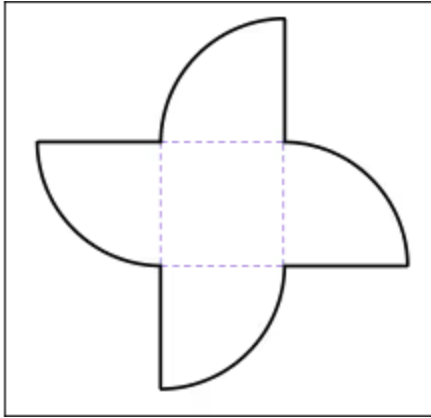


|   |                               |
|---|-------------------------------|
| a | value of $a$                  |
| b | radius of each quarter-circle |
| c | total length of line segments |
| d | total length both arcs        |
| e | total perimeter of shape      |

|   |           |
|---|-----------|
| c | 34 cm     |
| b | 8 cm      |
| e | 59.1 cm   |
| a | 9 cm      |
| d | $8\pi$ cm |



- 6 The square at the centre of this composite shape has an area of  $441 \text{ cm}^2$ . Find the perimeter of the composite shape in cm, rounded to the nearest integer. Fill in the blank



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